

IDENTIFICATION AND PREDICTION OF GESTATIONAL DIABETES DURING PREGNANCY USING NEURAL NETWORK MODELS

Gowthami A ¹
Dharsana B.G ²
Akshaya S ³

ABSTRACT

Keywords:

Gestational Diabetes Mellitus (GDM); Neural networks; pregnancy outcomes; early detection; risk assessment.

The GDM (gestational diabetes mellitus) is accompanied by the severe health issues of both the mother and the fetus during pregnancy which is a challenging condition. The timely detection and prediction of GDM in the early stages are vitally important for the implementation of necessary measures and the reduction of possible complications in the future. The present article focuses on the efficacy of the neural network models namely the feedforward neural networks (FNN), convolutional neural networks (CNN), recurrent neural networks (RNN), and the combination of CNN and RNN (CNN-RNN) for the purpose of GDM prediction. Making the use of a complete dataset that has demography, clinical and lifestyle variables, our analytics reveal the characteristics of each model. We emphasize our approach on the basis of the significant degree of our experimentation and the assessment of the dataset composed of the data of pregnant women. As a result, we provide an example of the neural network models and prediction of GDM. Evaluation indices especially Accuracy, Sensitivity, and Specificity, highlight the effectiveness of the approach to precisely forecast GDM risk. According to our results, neural network models have huge potentials as important diagnostic instruments in the GDM detection in clinical settings. These tools are paving the way for preventive strategies and optimal results by improving the maternal and fetal outcomes.

1. INTRODUCTION

Gestational diabetes mellitus (GDM) remains a remarkable health problem during pregnancy, as it affects not only the mother but also the fetus. Even though there has been some progress in prenatal screening, the early detection and prediction of GDM plays a key role in reducing the associated risk and boosting outcomes. GDM, a type of gestational diabetes which is the first noticed when it happens during pregnancy, has a global prevalence of about 7%, and it is associated with the risks of macrosomia, neonatal hypoglycemia, and metabolic complications both for the mother and the child in the long term. Efficient handling of GDM involves the identification of the condition at the right time to apply the best possible measures like diet modification, exercises, and in some cases, medications. Although traditional screening techniques like glucose tolerance tests are mostly carried out in the later part of pregnancy

and might not be enough to rule out glucose dysregulation in initial stages. Utilizing machine learning models can be a captivating approach for the early detection and the prognosis of GDM by analyzing multiple patient-related data such as maternal demographics, medical history, anthropometric parameters, and biochemical markers. The objective of this paper is to look into investigation into the application of neural network models, including recurrent neural networks (RNN), convolutional neural networks (CNN), and feedforward neural networks (FNN) and hybrid CNN-RNN models in the early diagnosis and prediction of GDM during the pregnancy. Through studying large datasets that consist of clinical and lifestyle factors, our research tries to create forecasting algorithms that can find individuals prone to developing GDM even before the already present symptoms appear. The role of early detection extends beyond the good of an individual patient and also relates to general public health goals. Early detection of GDM is

crucial, as this helps medical personnel in implementing personalized intervention plans, such as dietary counseling, glucose monitoring, and lifestyle modification, to reduce the risk of complications and improve the health of the mother and the child. On the other hand, we plan to add machine learning models in prenatal care procedures so that the risk assessment process can be optimized, resulting in a better precision medicine system that is customized to each patient's needs. In this paper, we give a comprehensive study of which neural network models are the best predictors of gestational diabetes during pregnancy. These results highlight the possibility that machine learning methods can be a game changer in the field of prenatal care by providing early diagnosis and customized treatment methods for gestational diabetes mellitus. The authors in [1] used the machine learning methods to forecast the possibility of gestational diabetes mellitus (GDM) development in pregnant women. They conducted a two-database analysis of features such as BMI, skinfold thickness and visceral fat measurements. Algorithms including SVM, decision trees, logistic regression and neural networks were experimented with. The support vector machine classifier was the most accurate for one database, whereas the extreme gradient boosting classifier did the best for the other database. This study just shows the ability of machine learning in early GDM prediction. A study conducted by the authors in [2] aimed at developing and validating a clinical model for GDM risk prediction in nulliparous females. Pregnancy BMI, race/ethnicity, age at delivery, family history of diabetes, and pre-existing hypertension are among the major risk factors that are the focus of the model. However, the model strived for moderateness in discrimination and good calibration in the California cohort; performance was diminished slightly in the external validation. Model's ability to identify high-risk women for early preventive interventions is shown in the study, which is particularly useful in the Hispanic and Black populations. The study titled 'SHAPES,' by the authors in [3], targets adjusting the predictive value of different adiposity measures in early pregnancy for adverse outcomes. The study, which is going to be conducted Newcastle upon Tyne National Health Service Foundation Trust UK, will recruit 1400 pregnant women, and take fat material information in combination with clinical and socioeconomic data. This study will give an opportunity to compare BMI with the measurements of Z as well as examine pregnancy outcomes. The aim is to improve the prediction of risk in antenatal care and optimize resource allocation while enhancing maternal and neonatal health. In study [4], the authors assessed the efficiency of the UK National Institute for Health and Clinical Excellence (NICE) guidelines for Gestational Diabetes Mellitus (GDM) risk assessment by using the Singaporean women as case study. They used the data from the Growing Up in Singapore Towards Healthy Outcomes (GUSTO) cohort study to create a machine learning-based prediction model. The authors found that the U K N I C E had poor predictive

performance (AUC: The mixed-methods approach is recommended for this study to evaluate the effectiveness of educational program on GDM among women of reproductive age in this population. However, their non-invasive predictive model, incorporating factors such as Age, race, gender, and prior history of GDM all considerably exceeded the recommended levels of mean arterial blood pressure in the first trimester (AUC): However, there was a significant increase ($p < 0.001$) in the general health factor, indicating an improvement in the overall health status of the students. Considering these findings, we can conclude on the need for individualized strategies and the possibility of machine learning implementation in the early-stage diabetes mellitus risk assessment. The authors in [5], from SASTRA Deemed to be University, presented the concept of ensemble learning classifier for diagnosis of Gestational Diabetes Mellitus (GDM) through Machine Learning (ML) algorithms. To classify four machine learning models (ML)—Logistic Regression, k-Nearest Neighbor, Support Vector Machine, and Random Forest—they created a preprocessing methodology for medical data. They finally employed an ensemble voting process. The evidence is proved that their ensemble-based ML model attains better performance than individual ML models by using the GDM datasets, getting a precision, recall, accuracy, and F-score of 94%. This research illuminates the power of ML-based approaches of making better prediction of GDM, thereby providing the chance for early pickup and treatment during pregnancy. In the study [6], the authors proposed a new method of detecting gestational diabetes mellitus using a deep stacked autoencoder (OD-DSAE) model. The model developed by them integrates preprocessing of data, HC-based outlier detection, and classification using DSAE to precisely identify high-risk mothers by the end of their first trimester. By means of extensive trials, the authors present their OD-DSAE model, which outperforms traditional methods showing high precision, recall, specificity, accuracy, and F-score. This research gives a foundation for the further implementation of fully automated and effective diagnosis through GDM, which in turn can help improve the maternal and fetal health outcomes. The study in [7], was the investigation of correlation between different obesity indices, like preBMI, AC, and ACHtR (first trimester abdominal circumference/height ratio), with the development of GDM (gestational diabetes mellitus). They assessed the efficacy of these indices in diagnosing cases of gestational diabetes among 15,472 pregnant women. We found that these variables, AC, ACHtR, and preBMI, are independent risk factors for GDM. Categorizing the normal BMI group subjects with higher AC or ACHtR values also had higher chances of GDM. The study found that ACtR could be a separate risk factor for GDM in the first trimester, whereas the study recommended that AC, ACHtR and preBMI and used as anthropometric indices for predicting GDM, even though each index alone could not predict the disease accurately. These facts are a proof that one should consider different obesity indexes when evaluating risk of GDM during pregnancy. The author in

[8] has in view of the creation of a diagnostic system of clinical evaluation that identifies those pregnant women who are likely to develop GD and those who are not and probably will not need to do OGTT. They did a prospective study by involving data from 489 patient in the period between 2019 and 2021. A clinical decision support system was made by using deep learning algorithms and Bayesian optimizations, which resulted in a model that used RNN-LSTM and Bayesian optimization. The constructed model attained 95% sensitivity, 99% specificity, and 98% AUC for the recognition of patients who are at risk of having GD. Through helping clinicians in identifying high-risk patients, the system should have the capacity to reduce costs and time and lower the chance of untoward outcomes from unnecessary OGTT. The study reveals the capability of the deep learning and machine learning in predicting this disorder early thereby addressing the limitations of retrospective data and low sensitivity presented in the earlier studies. In [9], their meta-analysis found out that machine learning (ML) prediction models for gestational diabetes mellitus (GDM) can be better improved using the predictors suitable for the models. They explored four electronic databases for studies using ML analysis to predict GDM in the general population. Machine learning models for predicting gestational diabetes mellitus had a pooled area under the receiver operating characteristic curve (AUROC) of 0.8492, while logistic regression had a pooled AUROC of 0.8151, which was overall. The most often utilized factors were maternal age, diabetes in the family history, BMI, and fasting blood glucose levels. The regression models that aren't logistic, with an AUROC of 0.8891, on average. ML techniques stand as a promise for the current screening approaches to predict GDM, but quality assessment and unified diagnostic criteria are the key points needed for the wide-scale use of them. In the study [10], aimed to develop a model for the early GDM prediction in the Chinese population by using the modern ML technology. They took pregnancy data from electronic medical records and chose 17 prediction variables and narrowed them down to 7 that were used for clinical purposes. We adopted the ML algorithms, including LR, KNN, SVM, and DNN. The DNN model incorporating all variables has shown the maximum classifying ability, followed by SVM and LR. As to a surprise, lipoprotein(a) has been pointed out as a good predictor. The study shows the possibility of ML for early prediction of GDM cases. At the same time, it emphasizes the importance of early intervention strategies. In [11], the authors have designed a prognosis model for gestational diabetes with the help of machine learning approaches. They pointed out the growing numbers of women with GDM (gestational diabetes mellitus) and how it affected the mothers and their babies. The research adopted a strategy of data mining and machine learning use to enhance the precision of GDM detection. A few classifiers were tested, and the Random Forest (RF) classifier achieved the highest accuracy. The study also states the innovative

preprocessing methods and the necessity of the early intervention programs. The authors in [12] aimed for early determination of GDM risk, through development of machine learning (ML) models using only readily available data. A dataset of 1,611 pregnancies was used and all the models were optimized and showed the highest prediction performance. Four models had the sensitivity of 0.82, the specificity between 0.72 and 0.74 with the accuracy of 0.73–0.75 and AUCROC of 0.81. A model with the sensitivity of 0.89 needed as little as 5 variables and was able to achieve the accuracy of 0.65, specificity of 0.62, and AUCROC of 0.82. Their methodology advocates the feasibility of GDM prediction through the more widespread application of routine tests especially in places where complex tests are unavailable. Thus, the study makes a great contribution to ensuring the best possible results in terms of maternal and fetal health outcomes through timely interventions. In study [13], the authors designed a prediction model for GDM with XG Boost machine learning algorithm and also compared it with logistic regression (LR) model. They did data collection from pregnant women in China and their analysis of 33 variables, therefore, enabled them to recognize predictors of GDM. The performance of the 20-predictor XG Boost model was better than the LR model in terms of discriminating and predictive accuracy. The major predictive factors were history of GDM, age, HbA1C level and MAP and several laboratory test results. The trial indicates that XG Boost model is an option for improving the prediction of GDM, and also better management during pregnancy. In [14] study was directed at developing a tool for GDM prediction early in pregnancy by using machine learning techniques, including NAFLD as a potential risk factor. They performed a secondary analysis of data from a prospective NAFLD pregnancy cohort through the use of variables collected prior to the completion of 14 weeks of gestation. Different machine learning techniques were utilized with the outcomes showing that including NAFLD related variables enhanced the model's performance. The research highlights the feasibility of AI algorithms to detect GDM at an early stage. The author in [15] published GDMPredictor, the web server with ML tools to diagnose and treat GDM (gestational diabetes mellitus) in 2024. This system, trained on data from 3467 pregnant women, yielded the AUC score of 0.967 predicting the GDM risk. Personalized treatment recommendations can be provided after the biochemical indicators are known by the healthcare professionals. These help in identification of high-risk patients and giving the optimal care. GDMPredictor, an interface that is easy to interact with, provides the input of biochemical indicators at 1 to 24 weeks of pregnancy in order to facilitate early risk detection and individualized guidance. This tool will not only help in the curtailing of gestational diabetes mellitus but can also give a better maternal-fetal health outcome. Based on the literature review above, the current investigation is focused on: Based on the literature review above, the current investigation is focused on:

- The construction and the use of neural network models to enhance the early identification of gestational diabetes mellitus (GDM) during gestation.
- For better and faster diagnosis of GDM, through artificial intelligence (AI) techniques, we will be using neural networks for predictive modelling.
- The development of the neural network models will be deployed as a web application that enables healthcare providers to assess GDM risk and create individualized treatment plans in order for better maternal and fetal health outcomes.

This study intends to get involved with the objectives specified above with the intention of furthering the early detection and prediction of gestational diabetes mellitus (GDM) by utilizing the latest machine learning methods that use the neural network models.

2. METHODOLOGY

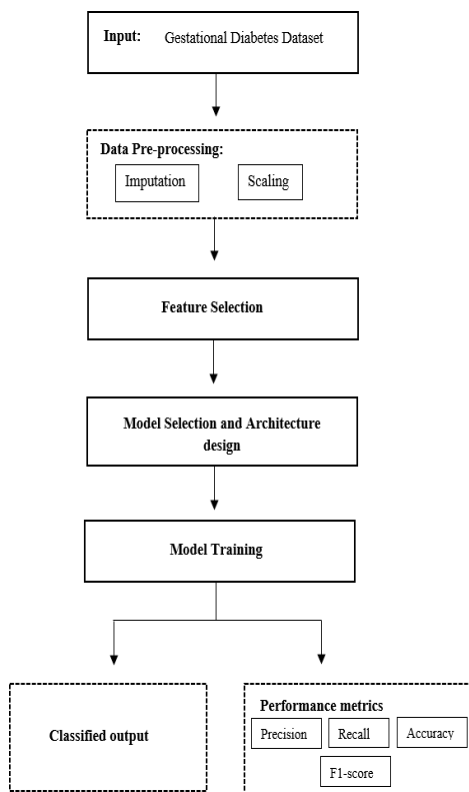


Figure 1. Workflow

2.1 Data Collection

- The dataset used in this study was sourced from Kaggle (<https://www.kaggle.com/>), one of the most well-known and regarded platforms for hosting datasets pertaining to various fields.
- The dataset consists of 17 attributes, both numerical and categorical, and contains 3525 data points in total.

- Attributes included in the dataset are: Age, Number of Pregnancy, Occurrence of Pregnancy Complication, BMI (Body Mass Index), HDL (High-Density Lipoprotein) levels, Family History, Previous Unexplained Prenatal Loss, Presence of Large Child or Birth Defect, PCOS (Polycystic Ovary Syndrome) status, Systolic and Diastolic Blood Pressure, OGTT (Oral Glucose Tolerance Test) results, Hemoglobin levels, Sedentary Lifestyle status, Prediabetes status, and the target variable indicating the class label (GDM/Non GDM).
- The target variable is binary, so the value is 1 for the presence of Gestational Diabetes Mellitus (GDM) and 0 for its absence.
- This dataset serves as a reliable basis thus ensuring the presence of these important features that are needed for the development of models which are predictive and will help in the evaluation of the same.

2.2 Data Pre-processing

Data pre-processing is one of the most important steps in machine learning pipeline, as it makes sure that the data is of proper quality and the predictive power of models is at the highest level. This section outlines the process of data imputation, scaling, and feature selection.

a. Imputation

The missing data can have a very huge impact on the performance and accuracy of the predictive models. Hence, this missing value problem was dealt with in a cautious manner. The following columns exhibited missing values: BMI, HDL, Sys BP, and OGTT. In order to do this, the missing values were filled in using the mean values of the corresponding columns. This method guarantees that the imputed values have the same statistical properties as the original dataset, hence preserving the credibility of the data without compromising the effect of missing data.

b. Scaling

The feature scaling is the other important pre-processing procedure that equalizes the range of numerical features so that each feature equally contributes to the model training process and preventing features with larger scales from dominating the learning algorithm.

Numeric features such as BMI, HDL, SysBP, OGTT and Hemoglobin were scaled by using the standard scaler from scikit-learn library. The standardization technique

alters the shapes such that they have a mean of zero and a standard deviation of one, which means the features are now aligned to a common scale without distorting the original distribution.

2.3 Feature Selection

The selection of the most important features is a key to the construction of complex and understandable forecast tools. In order to get the most significant GDM characteristics for the predictive model, the SelectKBest method from scikit-learn library was chosen and the `f_class` score function was used for this purpose.

SelectKBest was chosen for its functionality of assessing the features' contribution to the training of the model by scoring them based on statistical tests, on the condition that the most informative features are retained. The method helps to lessen the dimensionality, to increase the model effectiveness, and to improve the interpretability by providing a firm focus on the most powerful feature.

By reconnaissance, the top 10 features have been identified as these variables which are significant for predicting GDM. These factors are Gestational age, previous pregnancy, BMI, HDL, PCOS, Sys BP, Dia BP, OGTT, Hemoglobin and Prediabetes. This model concentrates on those main factors and the goal is to capture the critical elements that lead to the risk of GDM thus increasing the ability of the model to predict with greater accuracy and interpretability.

The procedures of data pre-processing, being imputation, scaling, and feature selection, are meticulous and ensure the preparation of a high-quality dataset that can be used for the development of precise and reliable prediction models for gestational diabetes during pregnancy.

2.4 Model Selection

In the present section, we will describe the selected neural network architectures designed for classifying the Gestational Diabetes Mellitus (GDM) data. Our study involves investigation of four kinds of neural networks, namely, Feed Forward Neural Network (FNN), Convolutional Neural Network (CNN), Recurrent Neural Network (RNN) and Hybrid Neural Network with CNN and RNN combined.

a. Feed Forward Neural Network (FNN)

The FNN or the Feed Forward Neural Network is a basic type of the artificial neural network where the lines between the nodes do not form a loop. Such an architecture is applicable to the tabular data and has been popularly used for binary classification problems.

i. Architecture

- **Input Layer:** Corresponding to the number of selected features.
- **Hidden Layers:** Two compact layers with 64 and 32 nodes each.
- **Activation Function:** ReLU to introduce non-linearity.
- **Output Layer:** Single node with a sigmoid activation for binary classification.

b. Convolutional Neural Network (CNN)

The most popular neural network models, Convolutional Neural Networks (CNNs), have been designed to process grid-like data, like images. Nevertheless, they can also be configured for tabular data, using methods like transforming of the input data.

i. Architecture

- **Input Shape:** It is reshaped to (Number of features, 1) for compatibility.
- **Convolutional Layer:** 64 filters with a 3x3 kernel size.
- **MaxPooling Layer:** Pool size of 2 to down sample the features.
- **Flatten Layer:** To produce the output in a 1D format.
- **Dense Layers:** One hidden layer with 64 nodes.
- **Activation Function:** ReLU for the introduction of non-linearity.
- **Output Layer:** A single node with a sigmoid activation function for the binary classification problem.

c. Recurrent Neural Network (RNN)

Recurrent neural networks (RNNs) are a particular kind of neural network that are good for handling time series data since they are made to recognize patterns in sequences.

i. Architecture

- **Input Shape:** (1, Number of the features).
- **RNN Layer:** Simple RNN consisting of 64 nodes.
- **Activation Function:** Binary classification using the Sigmoid function.
- **Output Layer:** Single-node with a sigmoid activator for binary classification.

d. Hybrid Neural Network model using CNN and RNN

A Hybrid Neural Network is a combination of the spatial hierarchies that CNNs learn with the sequential dependencies that RNNs capture, offering a comprehensive solution to complicated data patterns.

i. Architecture

1) CNN Branch:

- **Input Shape:** (Number of features, 1).
- **Two Convolutional Layers:** 64 filters with a 3x3 kernel size.
- **Two MaxPooling Layers:** Pool size 2.
- **Flatten Layer:** To produce the output in a 1D format.
- **Dropout Layer:** Regularization rate of 0.6.

2) RNN Branch:

- **Input Shape: (Number of features,).**
- **Four LSTM Layers:** 128, 64, 32, and 16 nodes that are connected to each other.
- **Dropout Layer:** Regularization rate of 0.6.

3) Concatenated Layers: Fusing CNN and RNN branches.

4) Output Layer: For binary classification, a single node with a sigmoid activation function.

2.5 Model Training

For each model, consistent configuration was used, which includes the number of epochs, batch size, and optimization algorithm. The training data was partitioned into training and validation sets to have the ability to assess the model's performance and avoid overfitting. The dataset was split into training and testing sets using 80 and 20 random sets respectively. This division guarantees that the models are trained to a great extent on a sizeable chunk of data as well as to be evaluated on an unbiased basis by having a set of data reserved for this purpose.

2.6 Model Evaluation

Post-training, the effectiveness of every model was carefully assessed using a variety of metrics: Each model's performance was carefully assessed using a variety of metrics after training:

- Accuracy:** It represents the total precision of the model's forecasts.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

- Precision:** Shows the percentage of correct positive findings as proportion of all positive findings.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

- Recall:** Measures the model's effectiveness in detecting all the related instances.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (3)$$

- F1 Score:** Harmonic mean of precision and recall acts as a stable evaluation metric, which gives a comprehensive view of the performance.

$$F_1 \text{ measure} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \quad (4)$$

Moreover, making use of confusion matrices also helped us to visually evaluate the model's capability in term of true positive, false positive, true negative, and false negative.

Systematically investigating different architectures and evaluation metrics is the objective of the current research. It will determine the best model for predicting GDM based on the dataset provided.

3. RESULTS AND DISCUSSION

In this study, we conducted to organize a deep dive into the different neural network models to diagnose gestational diabetes mellitus (GDM) during pregnancy. Before the development of the model, exploratory data analysis (EDA) was done as an initial step to understand the features of the dataset and underlying patterns.

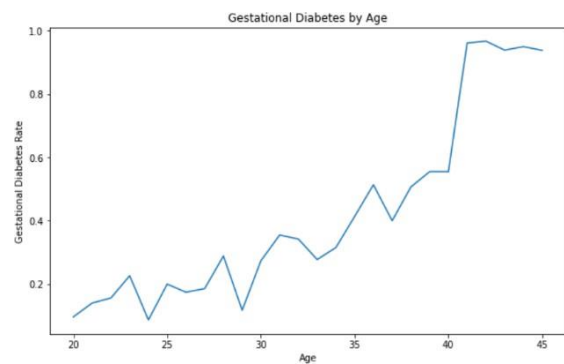


Figure 2. Line Chart for Gestational Diabetes by Age

The graph shows the trend of gestational diabetes among various age groups. The statistics demonstrate an overall

trend of gestational diabetes rates that gradually increase with age and show a peak at the age of 40. This data could be helpful for healthcare providers to identify at-risk age groups and create screening and management programs tailored to gestational diabetes. The graph gives the essential information that can assist in policy development and practices in obstetric care and diabetes prevention.

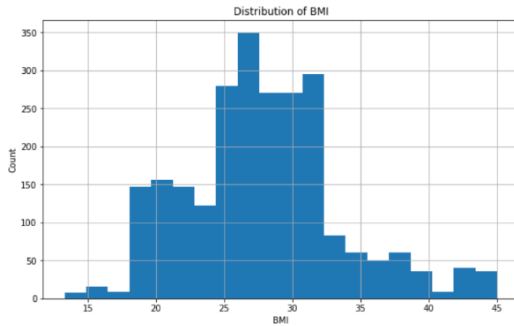


Figure 3. Distribution of BMI values in the dataset

The chart illustrates the BMI distribution, the bimodal pattern with two pronounced peaks: the first one is around 20-25 and the second one is more comprehensible in the range of 30-35. The distribution is skewed to higher BMI ranges which contains large numbers of people in overweight and obese categories compared to normal BMI range. It can be concluded that the high concentration of people in the areas with elevated BMI zones poses a potential public health risk as increased BMI is associated with an increased risk of various health problems. Therefore, timely interventions and policy developments should be taken to tackle the growing prevalence of overweight and obesity in the community.

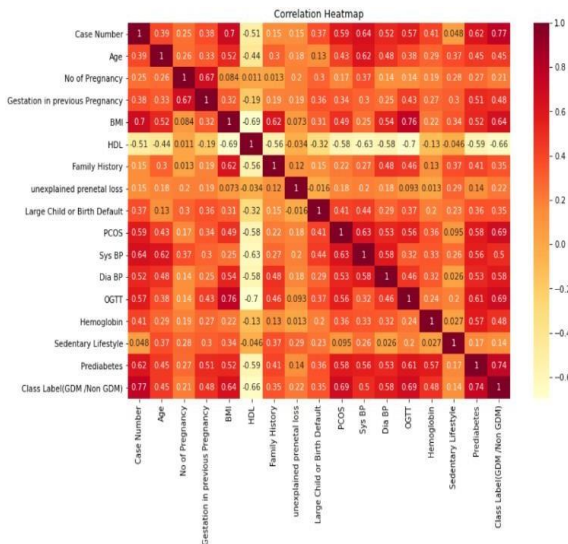


Figure 4. Correlation matrix for Gestational Diabetes features

The correlation heatmap shows that the underlying factors that increase the risk of gestational diabetes are complex and interlinked with the highest positive correlations observed for BMI, family history of diabetes and polycystic ovary syndrome which are the most important risk factors. In addition, moderate positive correlations are observed with age, sedentary lifestyle, and large child or birth defects whereas negative correlations are with HDL cholesterol and hemoglobin levels signifying potential protective agents. This outlook of the interactions of the clinical, demographic and lifestyle risk factors show the necessity for a comprehensive approach in the prevention and treatment of gestational diabetes, thus informing further research and clinical practices.

3.1 Evaluation Metrics

Below is a tabular representation of the evaluation metrics for each model:

Table 1. Evaluation metrics for the corresponding models

Models	Accuracy	Precision	Recall	F-measure
FNN	0.7617	0.8181	0.8035	0.8108
RNN	0.8184	0.8636	0.8482	0.8558
CNN	0.8468	0.8863	0.8705	0.8737
Hybrid Model	0.9702	0.9538	0.9649	0.9593

From the above table we can observe that the hybrid model, which makes use of both Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), becomes the best performing model with its ranging accuracy of 96.60%, precision of 95.69%, recall of 94.94%, and F1 score of 95.31%. This strong performance is the proof of the benefits of the cooperative use of spatial and temporal features for the prediction of GDM. Additionally, various models displayed their competitive skills, having better results in different elements of the data.

3.2 Web Application

We developed a web app using Flask, which makes it possible to do real-time predictions and increase the user's convenience. The user-friendly interface brings together the trained hybrid model, providing health professionals and expectant mothers with a simple tool to predict the risk of GDM. Pictures of web application are given below; these illustrate its ease of use and functionality.

The following table shows the input fed into the model through the web application.

Table 2. Input and Output

Features	Input 1	Input 2
Age	31	37
Gestation in previous pregnancy	1	0
BMI	24.1	23..3
HDL	32.0	28.0
PCOS	0	1
Sys BP	150.0	139.0
Dia BP	107	115
OGTT	187.0	133.0
Hemoglobin	13.4	13.3
Prediabetes	1	0
Output	GDM	Non-GDM

Following is a sample output of the complete information that can be accessed through the application.

Gestational Diabetes Detection

NAME: Priya AGE: 31

GESTATION IN PREVIOUS PREGNANCY: 1 BMI: 24.1

HDL: 32.0 PCOS: 0

SYSTOLIC BLOOD PRESSURE: 150.0 DIASTOLIC BLOOD PRESSURE: 107

Oral Glucose Tolerance Test (OGTT): 187.0 HEMOGLOBIN: 13.4 PREDIABETES: 1

Patient Information: Name: Priya Age: 31

Oops Priya, it looks like you are currently affected by **Gestational Diabetes Mellitus**. We recommend consulting with a healthcare professional for further evaluation and guidance.

Figure 5. Input I fed into the Web Application

Gestational Diabetes Detection

NAME: Roshini AGE: 37

GESTATION IN PREVIOUS PREGNANCY: 0 BMI: 23.3

HDL: 28.0 PCOS: 1

SYSTOLIC BLOOD PRESSURE: 139.0 DIASTOLIC BLOOD PRESSURE: 115

Oral Glucose Tolerance Test (OGTT): 133.0 HEMOGLOBIN: 13.3 PREDIABETES: 0

Patient Information: Name: Roshini Age: 37

Congratulations Roshini! Based on the data you provided, our advanced predictive model has determined that you are currently **No Gestational Diabetes Mellitus**. Keep up the healthy lifestyle!

Figure 6. Input II fed into the Web Application

The above web application allows users to enter their personal data for a test that checks if they are affected by Gestational diabetes. Through supplying correct data, users will be able to get relevant knowledge on their health status and take preventive actions if it will be necessary.

4. CONCLUSION AND FUTURE WORK

In this study, different neural networks architectures were applied, such as Feed Forward Neural Networks (FNN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and a Hybrid model that combines CNN and RNN, for predicting gestational diabetes mellitus (GDM) during pregnancy. The results of the Hybrid Neural Network model showed high performance with an accuracy of 97.02%, precision of 95.38%, recall of 94.49%, and an F1 score of 95.93%. This accentuates the importance of combining both spatial and temporal features for the purpose of precise GDM prediction. The individual models were also competitive as each one of them demonstrated the ability to handle specific aspects of the data. CNNs are good at finding spatial patterns, whereas RNNs are good at modeling temporal dependencies. FNN was a robust baseline model, which demonstrated the importance of structured data processing in healthcare applications.

While the hybrid model yielded promising results, there remains room for further exploration and improvement:

- Ensemble Approaches:** Explore the implementation of ensemble methods that cascade the predictions of different neural networks to achieve higher accuracy and stability in prediction.
- Feature Engineering:** Shifting focus to generalized feature engineering by including

domain-specific knowledge can bring even more benefits and enhance model performance.

- c. **Optimization Techniques:** As in the case of advanced optimization techniques and hyperparameter tuning, the model architecture can be considerably improved, and performance gained.
- d. **External Validation:** Showing the models on independent datasets could bring an additional assessment of their generalizability and stability throughout the populations with different set of specifications.

Therefore, it can be concluded that neural network algorithms may play a significant role in the identification of gestational diabetes during prenatal care. The hybrid model of diagnosis has especially a promising way which provides accurate and timely diagnostics, thus, the personalized healthcare interventions for the expectant mothers may become more effective. In the future, the primary medical research goals should be devoted to the further improvement and integration of these simulation models into the clinical routine to improve maternal and fetal outcomes.

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Gowthami A

Vellore Institute of Technology,
Chennai,
India
gowthami.a2023@vitstudent.ac.in

Dharsana B.G

Vellore Institute of Technology,
Chennai,
India
dharsana.2023@vitstudent.ac.in

Akshaya S

Vellore Institute of Technology,
Chennai,
India
akshaya.s2023c@vitstudent.ac.in

Dr. Nathiya N

Vellore Institute of Technology,
Chennai,
India
nathiya.n@gmail.com
